



IT582: Foundations of Machine Learning

Lab-6: Distance-Based Classification and K-Nearest Neighbors

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Submission Instructions

Upload **only one Jupyter Notebook (.ipynb)** file to Moodle.
The notebook should contain all code, outputs, plots and conclusions.

Name the notebook as:

[studentid]_lab6.ipynb

Implement the distance-based classifier and K-NN algorithm from scratch. Do not use built-in K-NN classifiers. Dataset-loading and evaluation utilities may be used.
Use a fixed random seed wherever randomization is required.

Task 1: Classification using Distance from Class Means

Generate a two-class dataset in $\mathbb{R}^2$ with

$$n_1 = n_2 = 100.$$

Generate Class 1 from

$$\mathbf{x} \sim \mathcal{N}\left(\begin{bmatrix} -1.5 \\ -1 \end{bmatrix}, \mathbf{I}\right)$$

and Class 2 from

$$\mathbf{x} \sim \mathcal{N}\left(\begin{bmatrix} 1.5 \\ 1 \end{bmatrix}, \mathbf{I}\right).$$

Randomly split the data into 70% training and 30% test sets.
Using only the training data, calculate the mean vector of each class:

$$\boldsymbol{\mu}_c = \frac{1}{n_c} \sum_{i:y^{(i)}=c} \mathbf{x}^{(i)}.$$

For a test point $\mathbf{x}$, calculate its Euclidean distance from each class mean,

$$D(\mathbf{x}, \boldsymbol{\mu}_c) = \|\mathbf{x} - \boldsymbol{\mu}_c\|_2,$$

and assign it to the class with the minimum distance:

$$\hat{y}(\mathbf{x}) = \arg \min_c D(\mathbf{x}, \boldsymbol{\mu}_c).$$

Report the test accuracy and plot:

- Training samples of both classes
- Estimated class means
- Decision boundary of the classifier

Task 2: K-Nearest Neighbors Classification

Use the same dataset from Task 1.

For a test point $\mathbf{x}$, calculate its distance from every training sample and select the K nearest observations.

Predict the class using majority voting among their labels.

Implement K-NN for

$$K \in \{1, 3, 5, 11, 21\}$$

using Euclidean distance.

For every value of K :

- Predict the labels of all test observations.
- Calculate the classification accuracy.

Plot classification accuracy versus K .

Also plot the decision boundaries for

$$K = 1, \quad K = 5, \quad K = 21.$$

Now generate Class 1 from

$$\mathbf{x} \sim \mathcal{N} \left(\begin{bmatrix} -1.0 \\ -0.5 \end{bmatrix}, \mathbf{I} \right)$$

and Class 2 from

$$\mathbf{x} \sim \mathcal{N}\left(\begin{bmatrix} 1.0 \\ 0.5 \end{bmatrix}, \mathbf{I}\right).$$

then repeat task 1 and 2 on this dataset.

Write observations and specify the algorithmic difference between the distance-from-means classifier and K-NN.

Task 3: K-NN Classification of MNIST

Use the **MNIST handwritten digit dataset** containing the classes

$$0, 1, \dots, 9.$$

Use the standard MNIST training and test sets. To keep the computation manageable, randomly select

$$10,000$$

training images and

$$2,000$$

test images using a fixed random seed.

Flatten each 28×28 image into a feature vector

$$\mathbf{x} \in \mathbb{R}^{784}$$

and scale pixel values to the range $[0, 1]$.

A. Effect of K

Using Euclidean distance, classify the test images using

$$K \in \{1, 3, 5, 7, 9\}.$$

For each value of K :

- Predict the test labels using majority voting.
- Report classification accuracy.

Plot test accuracy versus K and identify the value of K giving the highest accuracy.

B. Effect of Distance Function

The L_p distance between two feature vectors is

$$D_p(\mathbf{x}, \mathbf{z}) = \left(\sum_{j=1}^d |x_j - z_j|^p \right)^{1/p}.$$

For

$$p = 1,$$

this gives the Manhattan distance:

$$D_1(\mathbf{x}, \mathbf{z}) = \sum_{j=1}^d |x_j - z_j|.$$

For

$$p = 2,$$

this gives the Euclidean distance:

$$D_2(\mathbf{x}, \mathbf{z}) = \sqrt{\sum_{j=1}^d (x_j - z_j)^2}.$$

Using the same MNIST subset and a fixed value

$$K = 5,$$

compare K-NN classification using:

- Manhattan distance ($p = 1$)
- Euclidean distance ($p = 2$)

Report the classification accuracy for both distance functions.
Also display at least five test images along with:

- True digit
- Predicted digit using $p = 1$
- Predicted digit using $p = 2$

Write observations.